Student 14 **Jeyakannan L** 192525376RD 5 / 5 submitted

Q1. Program based on Classes and Objects

Score: 5.00 **TC: 3/3 passed**

CODE

```
import java.util.Scanner;
class Student{
    int rollNo;
    String name;
    double marks;
    void displayDetails(){
        System.out.println("Roll No: "+rollNo);
        System.out.println("Name: "+name);
        System.out.println("Marks: "+marks);
    }
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        Student s = new Student();
        s.rollNo = sc.nextInt();
        sc.nextLine();
        s.name = sc.nextLine();
        s.marks = sc.nextDouble();
        s.displayDetails();
        sc.close();
    }
}
```

INPUT

```
101
jk
75.93
```

OUTPUT

```
Roll No: 101
Name: jk
Marks: 75.93
```

Q2. Initialize Object Data Using Setter Methods

Score: 5.00 **TC: 3/3 passed**

CODE

```
import java.util.Scanner;
class Student{
    int rollNo;
    String name;
    double marks;
    public void setRollNo(int rollNo){
        this.rollNo=rollNo;
    }
    public void setName(String name){
        this.name=name;

    }
    public void setMarks(double marks){
        this.marks=marks;

    }
    public void displayDetails(){
        System.out.println("Roll No: "+ rollNo);
        System.out.println("Name: "+ name);
        System.out.println("Marks: "+ marks);
    }



    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        Student s = new Student();
        int rollNo=sc.nextInt();
        sc.nextLine();
        String name = sc.nextLine();
        double marks = sc.nextDouble();
        s.setRollNo(rollNo);
        s.setName(name);
        s.setMarks(marks);
        s.displayDetails();
        sc.close();
    }
}
```

INPUT

```
101
rahul
89.5
```

OUTPUT

```
Roll No: 101
Name: rahul
Marks: 89.5
```

Q3. Student Details Using Default and Parameterized Constructors

Score: 10.00 **TC: 3/3 passed**

CODE

```
import java.util.Scanner;
class Student{
    int rollNo;
    String name;
    double marks;
    Student(){
        rollNo = 101;
        name = "Rahul";
        marks=85.5;
    }
    Student(int rollNo, String name, double marks){
        this.rollNo=rollNo;
        this.name=name;
        this.marks=marks;

    }
    void displayDetails(){
        System.out.println("Roll No: "+ rollNo);
        System.out.println("Name: "+ name);
        System.out.println("Marks: "+marks);

    }
    public static void main(String[] args){
        Scanner sc=new Scanner(System.in);
        Student s1=new Student();
        s1.displayDetails();
        int rollNo=sc.nextInt();
        sc.nextLine();
        String name=sc.nextLine();
        double marks=sc.nextDouble();
        Student s2=new Student(rollNo,name,marks);
        s2.displayDetails();
        sc.close();



    }
}
```

INPUT

```
102
Arun
92.5
```

OUTPUT

```
Roll No: 101
Name: Rahul
Marks: 85.5
Roll No: 102
Name: Arun
Marks: 92.5
```

Q4. Net Salary Calculation Using Single Inheritance

Score: 10.00 **TC: 3/3 passed**

CODE

```
import java.util.Scanner;
class Employee{
    int empId;
    String empName;
    double basicSalary;

    void getEmployeeDetails(int empId, String empName, double basicSalary){
        this.empId = empId;
        this.empName = empName;
        this.basicSalary = basicSalary;
    }
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);

        int id = sc.nextInt();
        sc.nextLine();
        String name = sc.nextLine();
        double salary = sc.nextDouble();

        Salary s = new Salary();
        s.getEmployeeDetails(id, name, salary);
        s.calculateSalary();
        s.displayDetails();

        sc.close();
    }
}
class Salary extends Employee {
    double hra, da, netSalary;

    void calculateSalary(){
        hra = basicSalary * 0.20;
        da = basicSalary * 0.10;
        netSalary = basicSalary + hra + da;

    }
    void displayDetails(){
        System.out.println("Employee ID: " + empId);
        System.out.println("Employee Name: " + empName);
        System.out.println("Basic Salary: " + basicSalary);
        System.out.println("HRA: " + hra);
        System.out.println("DA: " + da);
        System.out.println("Net Salary: " + netSalary);

    }
}
```

INPUT

```
101
jeyakannan
500000
```

OUTPUT

```
Employee ID: 101
Employee Name: jeyakannan
Basic Salary: 500000.0
HRA: 100000.0
DA: 50000.0
Net Salary: 650000.0
```

Q5. Employee Gross Salary Calculation Using Multilevel Inheritance

Score: 10.00 **TC: 3/3 passed**

CODE

```
import java.util.Scanner;
class Employee
{
    int empId;
    String empName;
    void getEmployeeDetails(int id,String name)
    {
        empId=id;
        empName=name;

    }
    public static void main(String args[])
    {
        Scanner sc=new Scanner(System.in);
        int id=sc.nextInt();
        sc.nextLine();
        String name=sc.nextLine();
        double salary=sc.nextDouble();

        GrossSalary g=new GrossSalary();
        g.getEmployeeDetails(id,name);
        g.getBasicSalary(salary);
        g.calculateSalary();
        g.displayDetails();
    }
}
class Salary extends Employee
{
double basicSalary;
void getBasicSalary(double salary){
basicSalary=salary;
}
}
class GrossSalary extends Salary
{
double hra,da,grossSalary;
void calculateSalary()
{
hra=basicSalary*0.20;
da=basicSalary*0.10;
grossSalary=basicSalary+hra+da;
}
void displayDetails()
{
System.out.println("Employee ID: "+empId);
System.out.println("Employee Name: "+empName);
System.out.println("Basic Salary: "+basicSalary);
System.out.println("HRA: "+hra);
System.out.println("DA: "+da);
System.out.println("Gross Salary: "+grossSalary);
}
}
```

INPUT

```
101
Jk
4074
```

OUTPUT

```
Employee ID: 101
Employee Name: Jk
Basic Salary: 4074.0
HRA: 814.8000000000001
```

```
DA: 407.40000000000003
Gross Salary: 5296.2
```